

Last Name:

First Name:

Signature:

University of Waterloo, Mechatronics Engineering

SYDE252: Linear Systems and Signals
Dr. Matt Borland
Midterm Examination
Time Allowed: 120 minutes
Rooms: E7-3343, E7-3353

11:30am - 1:30pm, 19/06/2019

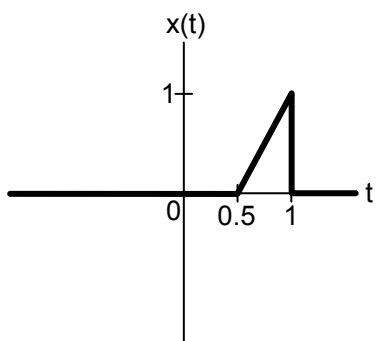
- *Just you, your brain, an approved calculator, and a pen!*
- *If you choose to write in pencil your test will NOT be eligible for regrading.*
- *Part marks can only be awarded for information that is clearly conveyed*
- *You are strongly advised to take the first 3-5 minutes of the test period to read the entire test over.*
- *An equation sheet has been provided as a separate page.*

Although this may be a most difficult thing, if one will do it, it can be done. There is nothing that one should suppose cannot be done.

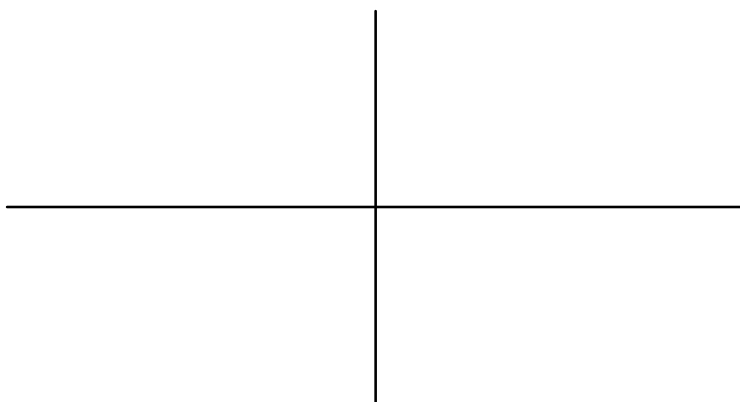
- *Yamamoto Tsunetomo, Hagakure*

Do not turn this page until you are told that you may do so.

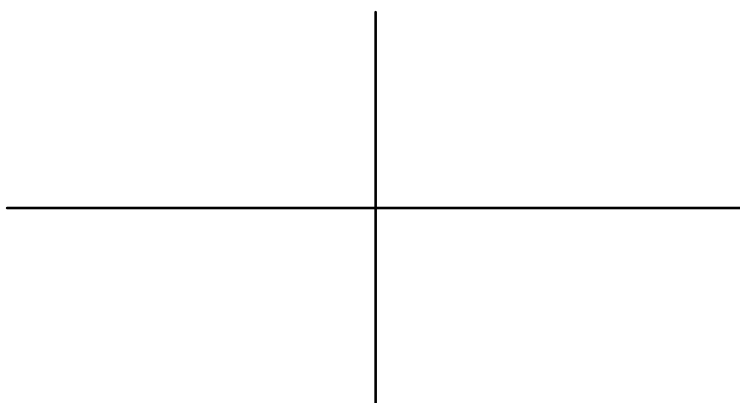
1. For the signal, $x(t)$, shown below, plot the following transformations. Be as complete as possible:



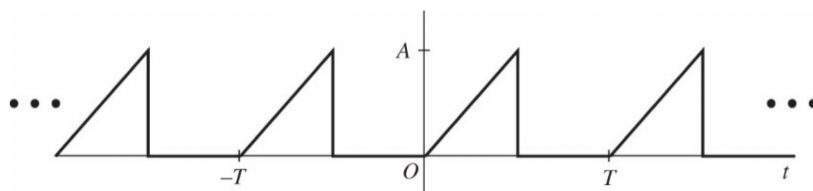
- (a) [1 mark] $x(-t - 0.5)$



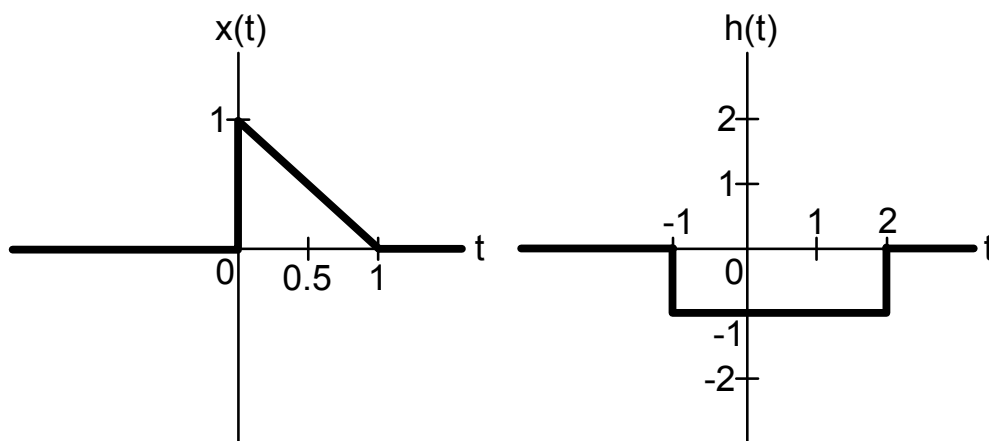
- (b) [1 mark] $x(1.5t - 2)$



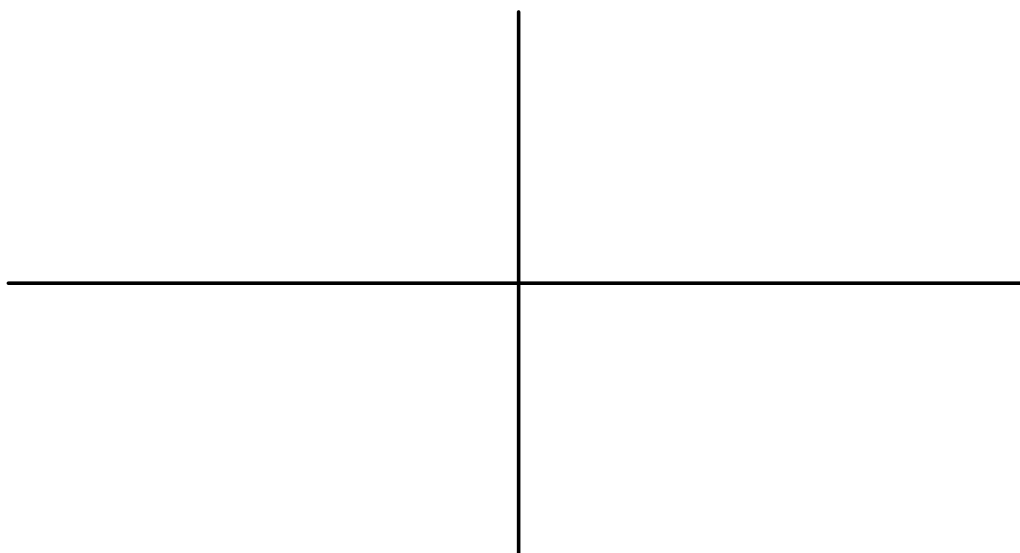
2. [3 marks] Determine the energy, power, and the RMS value of the 50% duty cycle dc-offset sawtooth wave $x(t)$ with peak amplitude A :



3. For the given signals, $x(t)$ and $h(t)$, use graphical convolution to calculate $y(t)$.



(a) [3 marks] Provide a sketch of $y(t)$ without developing equations or solving integrals:



4. For system defined by:

$$(D^2 + 4D + 3)y(t) = (D + 5)x(t)$$

(a) [2 marks] Find the Characteristic Equation and solve for its roots.

(b) [1 mark] Find the Characteristic Modes of the system.

...continues on next page...

(c) [4 marks] Find the Unit Impulse Response, $h(t)$, of the system.

5. Two inputs, temperature $T(t)$ and wind speed $V(t)$, produce an output, wind chill $W(t)$, according to:

$$W(t) = 35.7 + 0.6 T(t) - 35.7 V(t)^{0.16} + T(t)\{V(t)\}^{0.16}$$

...where the independent variable is t , time.

For each part below provide a proof or an argument to support your answer.

- (a) [1 mark] Is the system BIBO stable?

- (b) [1 mark] Is the system memoryless?

- (c) [1 mark] Is the system causal?

...continues on next page...

(d) [2 marks] Is the system linear?

...continues on next page...

(e) [2 marks] If the temperature is constant, that is $T(t) = k_T$, is the system linear?

6. Find the zero-state response to the input, $x(t)$, given the system's unit impulse response:

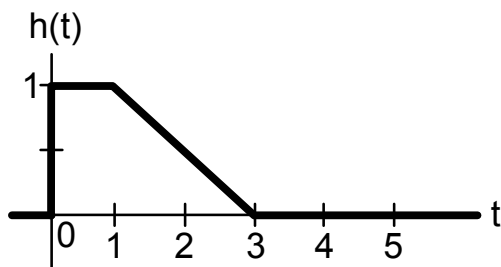
$$h(t) = [2e^{-3t} - e^{-2t}]u(t)$$

Do not use graphical convolution.

(a) [2 marks] $x(t) = e^{-t}u(t)$

(b) [2 marks] $x(t) = e^{-2t}u(t)$

7. A lowpass LTIC system has an impulse response, $h(t)$, shown below.



(a) [2 marks] Determine a rectangular approximation of $\hat{h}(t)$ to $h(t)$

(b) [1 mark] Determine the time constant, T_h , for $\hat{h}(t)$.

(c) [$\frac{1}{2}$ mark] Using $\hat{h}(t)$, what is the approximate radian cutoff frequency, ω_c , of this system?

(d) [$\frac{1}{2}$ mark] Assuming a frequency $\omega_0 \gg \omega_c$, what is the system response, $y(t)$, to the input $x(t) = \sin(\omega_0 t + 3.674\pi)$. Provide a numerical answer.

EXTRA PAGE FOR ROUGH WORK - Nothing on this page will be graded. Answer everything in the space provided in the questions.